

A. Zoom aware Tracking Implementation

Here we describe in detail our zoom aware reference tracking method.

A.1. Preliminary Equations

The Kalman Filter[8] is the optimal Bayes filter under the assumption of linear dynamics and Gaussian noise. As such, it is an iterative approach to estimate trajectory $\{\vec{x}_t\}_{t=1}^{\infty} \subseteq \mathbb{R}^n$ of true states, given corresponding *noisy* observations $\{\vec{z}_t\}_{t=1}^{\infty} \subseteq \mathbb{R}^m$, for some $n, m \in \mathbb{N}$. Suppose that $\{\vec{x}_t\}_{t=1}^{\infty}$ is Markov, and that

$$\mathbb{P}(\vec{x}_0) = \mathcal{N}(\vec{\mu}_0, \Sigma_0) \quad (1)$$

for some $\vec{\mu}_0 \in \mathbb{R}^n$ and $\Sigma_0 \in \mathbb{R}^{n \times n}$. Further suppose that for all $t = 1, 2, \dots$:

$$\mathbb{P}(\vec{x}_t | \vec{x}_{t-1}) = \mathcal{N}(A_t \vec{x}_{t-1}, R_t) \quad (2)$$

$$\mathbb{P}(\vec{z}_t | \vec{x}_t) = \mathcal{N}(C_t \vec{x}_t, Q_t) \quad (3)$$

for some $A_t, R_t \in \mathbb{R}^{n \times n}$ and $C_t \in \mathbb{R}^{m \times n}$. Under these assumptions, the Kalman Filter gives us the following iterative equation set for each $t = 1, 2, \dots$ [10]:

$$\vec{\mu}_t = \mathbb{E}(\vec{x}_t | \vec{z}_{1:t-1}) = A_t \vec{\mu}_{t-1} \quad (4)$$

$$\bar{\Sigma}_t = \text{Var}(\vec{x}_t | \vec{z}_{1:t-1}) = A_t \Sigma_{t-1} A_t^T + R_t \quad (5)$$

$$K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + Q_t)^{-1} \quad (6)$$

$$\vec{\mu}_t = \mathbb{E}(\vec{x}_t | \vec{z}_{1:t}) = \vec{\mu}_t + K_t (\vec{z}_t - C_t \vec{\mu}_t) \quad (7)$$

$$\Sigma_t = \text{Var}(\vec{x}_t | \vec{z}_{1:t}) = (I - K_t C_t) \bar{\Sigma}_t \quad (8)$$

As Gaussian distributions are maximised at their mean, this gives a convenient closed form to iteratively estimate the state of a trajectory based on observations. The Markov assumption central to the Kalman Filter is quite strong, but it has been observed in practice that this procedure is robust to thoughtful violations [10].

The Kalman Filter is a powerful tool for predicting the position of a designated target. However, in practice the challenge is often assigning observations to one or multiple simultaneous tracks. Concretely for this paper: We wish to track multiple objects within a scene, and must correctly associate detections in a given frame with the appropriate trajectory. Joint Probabilistic Data Association (JPDA), introduced in [1] and later corrected by [4], has become a standard approach to this problem.

On any given time-step, suppose we receive $q \in \mathbb{N}$ observations $Z = \{\vec{z}_i\}_{i=1}^q$ to associate with $p \in \mathbb{N}$ trajectory prior states $X = \{\vec{x}_i\}_{i=1}^p$ (Note we omit the time-step information for clarity as the following procedure is independent of time-step). For each $i = 1, \dots, p$, define $A^{(i,0)}$ as the event that no observation corresponds to state i , and define $A^{(i,j)}$ for each $j = 1, \dots, q$ as the event that

observation j corresponds to state i . A joint association $A_{\text{joint}} = \{A^{(1,j_1)}, A^{(2,j_2)}, \dots, A^{(p,j_p)}\}$ is feasible if for each $s = 1, \dots, p$, $j_p \in \{1, \dots, p\}$ and $\forall r \in 1, \dots, p$, $s \neq r \implies j_s \neq j_r \vee j_r = 0$, which is to say that each observation may be associated with at most one state.

We assume that observations (drawn from $\mathbb{R}^2$ in our case) are normalised to the unit square, that each state may manifest at most one true observation per time-step, and that any observation not associated with a state is solely noise. Furthermore, we assume that noisy detections were generated independently and uniformly and distributed, with the number of noisy measurements drawn from a $Poiss(\lambda)$ distribution, where $\lambda \in [0, 1]$ is a hyperparameter describing noise density. Finally, we assume that each state $i = 1, 2, \dots, p$ had probability $P_{\text{det}}^{(i)} \in [0, 1]$ of manifesting an observation. Under these assumptions, JPDA yields the probability of a feasible joint association to be [4]:

$$\mathbb{P}(A_{\text{joint}} | X, Z) = \frac{\lambda^{|\{A^{(i,j)} \in A_{\text{joint}} \text{ s.t. } j \neq 0\}|}}{c} \cdot \left(\prod_{A^{(i,j)} \in A_{\text{joint}} \text{ s.t. } j \neq 0} P_{\text{det}}^{(i)} \mathbb{P}(\vec{z}_j | \vec{x}_i) \right) \cdot \left(\prod_{A^{(i,j)} \in A_{\text{joint}} \text{ s.t. } j = 0} (1 - P_{\text{det}}^{(i)}) \right) \quad (9)$$

where c is some normalising constant. The probability of an infeasible joint association is taken to be 0. Using the law of total probability, individual assignment probabilities can then be extracted from a selected joint association to compute posterior trajectory states:

$$\mathbb{P}(A^{(i,j)} | X, Z) = \begin{cases} \sum_{A_{\text{joint}} \ni A^{(i,j)}} \mathbb{P}(A_{\text{joint}} | X, Z), & j \neq 0 \\ 1 - \sum_{k=1}^q \mathbb{P}(A^{(i,k)} | X, Z), & j = 0 \end{cases} \quad (10)$$

Since the introduction of JPDA, more sophisticated statistical models of the multi-target tracking problem have been developed, including some that directly model track birth and death such as Poisson Multi-Bernouli Mixtures [5]. However such methods incur significant complexity penalties, impose stronger assumptions on models, and are designed with prolonged continuous tracking in mind. JPDA remains a staple approach for simple scenarios where performance is necessary, with modern works such as [7] and [11] developing efficient implementations of the JPDA method.

A.2. Tracking Implementation

Before we begin, it is important to note that while not relevant to our description within the main text, we normalised our coordinate system to the unit square for modelling. The reader should consider our coordinate system normalised for the remainder of this section. Under this normalisation, the region zoomed in on was defined by the square:

$$\vec{\gamma} = (\vec{\alpha}, \theta) \in \mathbb{R}^3 \quad (11)$$

where $\vec{\alpha} = (\alpha_x, \alpha_y) \in \mathbb{R}^2$ denotes the center of the zoom region, and θ denotes its radius (half side length).

We assumed that the motion of a detected vessel on the screen was one of constant (likely 0) acceleration over the relatively short time-frame we collected, with Gaussian noise in both position and observation. This assumption held particularly well for vessel detection as sudden changes in direction are difficult for watercraft. We modelled a detection's true state at time $t = 0, 1, \dots$ to be of the form:

$$\vec{x}_t = (p, q, p', q', p'', q'') \in \mathbb{R}^6, \quad (12)$$

where $(p, q) \in \mathbb{R}^2$ denotes the position of the detection's centre, $(p', q') \in \mathbb{R}^2$ denotes the velocity of this centre, and $(p'', q'') \in \mathbb{R}^2$ denotes the acceleration of this centre. Observations were reduced to the coordinates of a detection's center with form:

$$\vec{z}_t = (p, q) \in \mathbb{R}^2 \quad (13)$$

Assuming a constant (but uncertain) velocity, we modelled the state of a detection at time $t = 1, 2, \dots$ as:

$$\mathbb{P}(\vec{x}_t | \vec{x}_{t-1}) = \mathcal{N} \left(\begin{bmatrix} I_3 & \delta_t I_3 \\ 0_{3 \times 3} & I_3 \end{bmatrix} \vec{x}_{t-1}, \rho \begin{bmatrix} \frac{\delta_t^3}{3} I_3 & \frac{\delta_t^2}{2} I_3 \\ \frac{\delta_t^2}{2} I_3 & \delta_t I_3 \end{bmatrix} \right) \quad (14)$$

$$\mathbb{P}(\vec{z}_t | \vec{x}_t) = \mathcal{N}([I_3 \quad 0_{3 \times 3}] \vec{x}_t, \sigma_{meas}^{*2} I_3) \quad (15)$$

where $\delta_t \in \mathbb{R}$ simply denotes the true difference in timestamp between steps $t-1$ and t ; $\rho \in [0, 1]$ is a hyperparameter defining the falloff in uncertainty with subsequent measurements. Due to the coordinate transformation explained in the body text, we have that:

$$\sigma_{meas}^* = \begin{cases} \sigma_{meas}, & \text{Prior to Zoom Boundary} \\ 2\theta\sigma_{meas}, & \text{After Zoom Boundary} \end{cases} \quad (16)$$

where $\sigma_{meas} \in \mathbb{R}$ is a hyperparameter denoting the standard deviation in measurement. Experimentally We found

$$\rho = 0.0005 \quad (17)$$

$$\sigma_{meas} = 0.01 \quad (18)$$

To be sufficient for our application.

Under our motion assumptions, a Kalman Filter was applied to the motion of known objects on screen. We used an efficient implementation of JPDA formulated by [11] to associate detections on each frame with the set of known trajectories. This implementation was selected as data association was a complexity bottleneck and our task prioritised speed over absolute precision. Experimentally We found JPDA hyperparameters

$$\lambda = 0.1 \quad (19)$$

$$P_{det}^{(*)} = 0.75 \quad (20)$$

were sufficient for our application.

On a given frame with $D \in \mathbb{N}$ detections, Trajectory $i \in \mathbb{N}$ was not assigned an observation if

$$\arg \max_{j=0,1,\dots,D} \mathbb{P}(A^{(i,j)} | X, Z) = 0 \quad (21)$$

where $\mathbb{P}(A^{(i,j)} | X, Z)$ is defined in Equation 10. In the event a trajectory was not assigned an observation, its updated state was taken to be the prior belief defined by Equation 4 and Equation 5. This increases covariance compared to the case where an observation was made, representing the increased uncertainty.

We assumed that there would be a constant but unknown number of objects across the tracking period. Object discovery was handled using a system inspired by SORT[2]. On each frame, after assigning detections to trajectories, a new tentative trajectory was constructed for each detection left unassigned. These tentative trajectories were initialised with initial state:

$$\vec{x}_0 \sim \mathcal{N} \left((p_d, q_d, \vec{0}_4), \begin{bmatrix} \sigma_{pos}^2 I_2 & 0_{2 \times 2} & 0_{2 \times 2} \\ 0_{2 \times 2} & \sigma_{vel}^2 I_2 & 0_{2 \times 2} \\ 0_{2 \times 2} & 0_{2 \times 2} & \sigma_{acc}^2 I_2 \end{bmatrix} \right) \quad (22)$$

where $(p_d, q_d) \in \mathbb{R}^2$ was the centre of the inciting detection; and σ_{pos} , σ_{vel} , and σ_{acc} were hyperparameters denoting the standard deviation in position, velocity and acceleration respectively. Experimentally we found

$$\sigma_{pos} = \sigma_{vel} = \sigma_{acc} = 0.001 \quad (23)$$

to be sufficient for our application. Due to the change of coordinates After assigning detections to known trajectories on frame t^1 , before new tentative tracks were initialised, each tentative trajectory greedily claimed the unassociated observation closest to its prior belief according to squared Mahalanobis distance[9]:

$$\delta_{mah}^2 = ([I_{2 \times 2} \quad 0_{2 \times 2}] \vec{\mu}_t - \vec{z}) \cdot (\bar{\Sigma} + \sigma_{meas}^2 I_{2 \times 2})_t^{-1} \cdot ([I_{2 \times 2} \quad 0_{2 \times 2}] \vec{\mu}_t - \vec{z})^T \quad (24)$$

¹We suppress notation for clarity, but formally each trajectory independently measures time-steps from its inception

where $\vec{\mu}_t$ and $\vec{\Sigma}_t$ are given in Equation 4 and Equation 5 respectively, and $\vec{z} \in \mathbb{R}^2$ is the observation being measured to. Mahalanobis distance was used to account for the (relatively) large uncertainty in position and velocity for a new trajectory. As the prior belief was normally distributed, the squared Mahalanobis distance from it to an observation had χ^2 distribution with 2 degrees of freedom[9]. Using this, we gated observation selection by requiring the claimed point lie within the corresponding 95% confidence interval (that is, possessed a squared Mahalanobis distance under ≈ 5.991 [9]). If the closest point did not satisfy this condition, then no observation was assigned to the tentative trajectory on that frame. On such frames, the state of the tentative trajectory was updated to be its prior prediction as with confirmed trajectories. Once a tentative trajectory successfully claimed an observation over four separate frames (six if it was discovered after zooming), it was promoted to a complete known trajectory to take part in JPDA. If a tentative trajectory failed to claim an observation across three separate frames before promotion, it was dropped as noise. This system limited the influence of noise on the joint distribution for association.

An important note specific to zoom aware tracking: The Mahalanobis based tentative association method was used for *all* tracks on the first zoom frame, as the assumptions of JPDA were sufficiently violated.

A.3. Homography Sanity Checks

As part of projecting zoomed detections to the unzoomed coordinate system in the main text, we compute Homography:

$$H = \begin{bmatrix} h_{11} & h_{12} & 0 & 0 & h_{13} \\ h_{21} & h_{22} & 0 & 0 & h_{23} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ h_{31} & h_{32} & 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{5 \times 5} \quad (25)$$

This Homography ignores bounding box scale, and thus for simplicity we consider Homography

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3} \quad (26)$$

within this section. This Homography has 8 Degrees of Freedom, and thus can be solved by 4 point correspondences. We applied the RANSAC based algorithm presented in [6] (Algorithm 4.6) to rapidly and robustly determine the least median squares estimate of H . However, this means that H is ultimately fit to noisy data, and thus the numerical solution may be nonsense geometrically. As such, before we accept H , we check that the following geometric sanity checks hold. If H fails any of these checks, we take I_3 instead.

RANSAC Inlier Ratio We first check:

$$\frac{N_{in}}{N} < \tau_{Inlier} \quad (27)$$

Where $N \in \mathbb{N}$ is the number of point correspondences used to compute H , $N_{in} \in \mathbb{N}$ is the number of point correspondences in agreement with H , and $\tau_{Inlier} \in \mathbb{R}$ is the decision threshold. This guards against H being the result of spurious correspondences. We found $\tau_{Inlier} = 0.6$ to be sufficient for our experiments.

Projective Distortion As we assume that we are observing a plane, the apparent scale of objects should not change significantly, so we check:

$$\|(h_{31}, h_{32})^T\|_2 < \tau_{Perspective} \quad (28)$$

Where $\tau_{Perspective} \in \mathbb{R}$ is the decision threshold. We found $\tau_{Perspective} = 0.05$ to be sufficient for our experiments.

Homogeneous Scale We also guard against homogenous coordinates being sent to infinity by a near zero denominator [6] by checking the extreme values (The corners of the unit square):

$$|h_{31} + 1| > \tau_{Denom} \quad (29)$$

$$|h_{32} + 1| > \tau_{Denom} \quad (30)$$

$$|h_{31} + h_{32} + 1| > \tau_{Denom} \quad (31)$$

Where $\tau_{Denom} \in \mathbb{R}$ is the decision threshold. We found $\tau_{Denom} = 0.25$ to be sufficient for our experiments.

Global Displacement We check that H has not dramatically translated the coordinate system. We do this again by checking the extreme values. Define the transformed unit square as

$$V = (\vec{v}_{(0,0)}, \vec{v}_{(0,1)}, \vec{v}_{(1,0)}, \vec{v}_{(1,1)}) \quad (32)$$

$$\vec{v}_{(0,0)} = \begin{pmatrix} h_{13} & h_{23} \\ h_{33} & h_{33} \end{pmatrix} \quad (33)$$

$$\vec{v}_{(1,0)} = \begin{pmatrix} \frac{h_{11} + h_{13}}{h_{33}} & \frac{h_{21} + h_{23}}{h_{33}} \end{pmatrix} \quad (34)$$

$$\vec{v}_{(0,1)} = \begin{pmatrix} \frac{h_{12} + h_{13}}{h_{33}} & \frac{h_{22} + h_{23}}{h_{33}} \end{pmatrix} \quad (35)$$

$$\vec{v}_{(1,1)} = \begin{pmatrix} \frac{h_{11} + h_{12} + h_{13}}{h_{33}} & \frac{h_{21} + h_{22} + h_{23}}{h_{33}} \end{pmatrix} \quad (36)$$

We Check:

$$\|\vec{v}_{(0,0)}\|_2 < \tau_{Displacement} \quad (37)$$

$$\|\vec{v}_{(0,1)} - (0, 1)\|_2 < \tau_{Displacement} \quad (38)$$

$$\|\vec{v}_{(1,0)} - (1, 0)\|_2 < \tau_{Displacement} \quad (39)$$

$$\|\vec{v}_{(1,1)} - (1, 1)\|_2 < \tau_{Displacement} \quad (40)$$

Where $\tau_{Displacement} \in \mathbb{R}$ is the decision threshold. We found $\tau_{Displacement} = 0.1$ to be sufficient for our experiments.

Global Shear We check that H has not dramatically sheared the coordinate system. Given the transformed unit square defined in Equation 32, its area is:

$$A_V = \frac{1}{2} \left| (\vec{v}_{(0,0)} \times \vec{v}_{(0,1)}) + (\vec{v}_{(0,1)} \times \vec{v}_{(1,0)}) + (\vec{v}_{(1,0)} \times \vec{v}_{(1,1)}) + (\vec{v}_{(1,1)} \times \vec{v}_{(0,0)}) \right| \quad (41)$$

Given the area of the unit square is 1, it is straightforward to check that the ratio of areas is consistent:

$$\tau_{Area} < A_V < \tau'_{Area} \quad (42)$$

Where $\tau_{Area}, \tau'_{Area} \in \mathbb{R}$ are the decision thresholds. We found $\tau_{Area} = 0.8$ and $\tau'_{Area} = 1.2$ to be sufficient for our experiments.

Anisotropy Finally, we check that H has not dramatically stretched the coordinate system by sampling the center of the transformed region (the point $(\frac{1}{2}, \frac{1}{2})$).

Consider the transformed centerpoint:

$$\vec{v}_c = (v_{cx}, v_{cy}) \quad (43)$$

$$v_{cx} = \frac{h_{11}}{2h_{33}} + \frac{h_{12}}{2h_{33}} + \frac{h_{13}}{h_{33}} \quad (44)$$

$$v_{cy} = \frac{h_{21}}{2h_{33}} + \frac{h_{22}}{2h_{33}} + \frac{h_{23}}{h_{33}} \quad (45)$$

The Jacobian of H around this center point is:

$$J_H(\vec{v}_c) = \frac{1}{\frac{h_{31}}{2} + \frac{h_{32}}{2} + h_{33}} \begin{bmatrix} h_{11} - h_{31}v_{cx} & h_{12} - h_{32}v_{cx} \\ h_{21} - h_{31}v_{cy} & h_{22} - h_{32}v_{cy} \end{bmatrix} \quad (46)$$

And its singular value decomposition is given by:

$$J_H(\vec{v}_c) = U \begin{bmatrix} e_1 & 00 & e_2 \end{bmatrix} V^T \quad (47)$$

For some affine rotations $U, V \in \mathbb{R}^{2 \times 2}$ [3]. e_1 describes the most significant scale component, and e_2 the least [3]. This means we wish to check these components are balanced to minimise stretching:

$$\frac{e_1}{e_2} < \tau_{Anisotropy} \quad (48)$$

Where $\tau_{Anisotropy} \in \mathbb{R}$ is the decision threshold. We found $\tau_{Anisotropy} = 1.5$ to be sufficient for our experiments.

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